A Chimera method for thermal part-scale metal Additive Manufacturing simulation

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Abstract

This paper presents a Chimera approach for the thermal problems in welding and metallic Additive Manufacturing (AM). In particular, a moving mesh is attached to the moving heat source while a fixed background mesh covers the rest of the computational domain. The thermal field of the moving mesh is solved in the heat source reference frame. The chosen framework to couple the solutions on both meshes is a non-overlapping Domain Decomposition (DD) with Neumann-Dirichlet transmission conditions.

Increased steadiness and accuracy within the vicinity of the Heat Affected Zone (HAZ) are the main advantages of this approach. The steadiness gain allows for the use of larger time steps, which is vital in AM applications and, in particular, Laser Powder Bed Fusion (LPBF), where the disparity of time scales represents a major hurdle. Moreover, enhanced accuracy can be observed in the resulting morphology of the melt pool. It will be shown that the method addresses classical shortcomings pointed out by Goldak without requiring the use of an asymmetrical heat source profile.

1. Introduction

Thermal part-scale simulation of metallic AM is necessary to analyze microstructure characteristics and detect regions with high thermal gradients (Lu et al., 2022; Fu and Guo, 2014). However, this is a daunting task, often implying long execution times and high computational costs. Modelling these problems is challenging due to their "extremely multiscale" nature (Hodge, 2021). In these applications, builds of characteristic size L_{print} are driven by a laser of characteristic size L_{hs}, where the ratio L_{print}/L_{hs} is about $100 - 1000$.

The discrepancy of temporal scales is even more marked. The timescale of the heat source, T_{hs}, can be estimated as the time required by the laser to travel one radius R, $T_{hs} = R/V$, with V the speed of the heat source. T_{hs} will be used as a unit of time in the remainder of this work. In LPBF, it is in the order of 10^{-4} seconds. This is prohibitively small compared to the *net printing time*, the total time the laser is turned on. The ratio T_{print}/T_{hs} can be orders of magnitude bigger than 10^5.

In practice, the timescale T_{hs} dictates the following constraint on the time-step Δt :

$$\Delta t \leq T_{hs} \tag{1}$$

If inequality (1) is not respected, the numerical solution will display a spotty pattern: elements are skipped over resulting in numerous erroneous temperature peaks as can be observed on fig. 1.2. Many authors opt for $\Delta t = 0.5\ T_{hs}$ (Patil et al., 2021; Michaleris, 2014; Stender et al., 2018; Proell et al., 2024) and commercial software packages like `Netfabb` do not allow for a time-step bigger than a T_{hs}[1]. For sizeable builds, this constraint is too restrictive and renders "naive modelling" [2] infeasible: Hodge (2021) estimates

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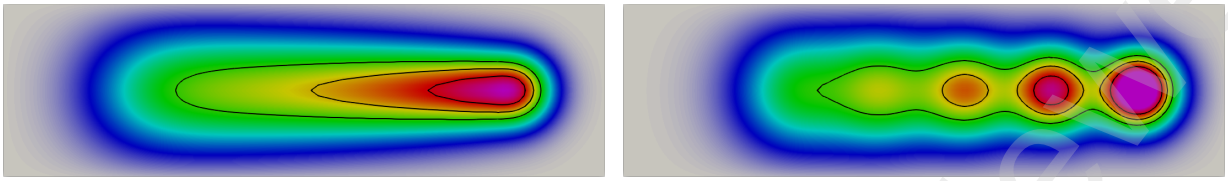
[1] `https://help.autodesk.com/view/NETF/2021/ENU/?guid=GUID-67A6D8F0-AA04-436B-9FAF-A912BE00B497`

[2] As in comprehensive first-principles modelling. Wording from Hodge (2021); King et al. (2015).

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that the simulation of a 19.2 mm cube printed using typical LPBF parameters would take $\mathcal{O}(10^8)$ time-steps. In the quest to obtain results within a reasonable time interval (in the order of days) for builds of practical interest, a time-step should span tens of microseconds at most (Proell et al., 2024).



(1.1) $\Delta t = 1T_{hs}$ (1.2) $\Delta t = 2T_{hs}$

Figure 1: 2D heating track example with admissible and inadmissible time-step sizes according to inequality (1).

On the one hand, the contrast of spatial scales has been amply studied and is typically addressed through Adaptive Mesh Refinement (AMR) (Moreira et al., 2022; Ganeriwala et al., 2021; Soldner and Mergheim, 2019). On the other hand, the disparity of time scales has only recently picked up the interest of researchers and there does not exist a standard solution yet. There are 2 classes of solutions in the literature to reduce the number of time-steps of the whole structure: *lumped heat sources* and *substepping*, the former being the most popular of the two.

Replacing the accurate heat source profile by an averaged source term like in Chiumenti et al. (2017); Malmelöv et al. (2019); Soldner and Mergheim (2019) allows bypassing the restriction of the heat source timescale to advance hatch-by-hatch or even layer-by-layer at the cost of accuracy. Van Elsen et al. (2007) shows that although the temperature field far from the HAZ is matched, the size of the melt pool as well as the peak temperature are underestimated. One can compensate by numerically calibrating other problem parameters as is done in *flash heating* (Patil et al., 2021): the deposition speed is artificially increased in order to obtain more faithful peak temperatures.

As an alternative, *substepping* does not necessarily compromise the accuracy of the results. Rather than taking uniform time-steps for the whole structure, finer time-steps are taken close to the heat source and coarser time-steps far from it. Puso and Hodge (2023) showcases speedup of factors 20-90 as their numerical example is scaled up. This is achieved through a "multirate integration scheme" (Hodge, 2021) so that the degrees of freedom (DOFs) are spatially partitioned into fast and slow. A time-step is then subdivided into (1) a global predictor step using a large Δt, (2) corrector steps using a small Δt for the fast DOFs and (3) a final corrector step for all DOFs. Viguerie et al. (2022) uses a similar predictor-corrector scheme together with the overlapping DD framework approach described in Carraturo et al. (2021). They obtain a computational speedup of 2.44 in a 3D welding example[3]. The work of Soldner and Mergheim (2019) showcases a DD framework with heterogeneous time-steps combined with a lumped heat source. Kopp et al. (2022) is able to naturally address both the spatial and temporal multiscale natures of their welding examples through space-time FEM: the time interval is meshed and its (de)refinement is handled with AMR.

There also exist techniques to reduce the computational cost of a single time-step. Matrix-free FEM (Dugast et al., 2021) and CPU vectorization (Proell et al., 2024) can enhance performance. Explicit time-stepping can also be used as a speedup technique: although it brings up stability concerns, the critical time-step is less restrictive than constraint (1) by a factor of $5-10$ in the examples of Essongue et al. (2022).

The model proposed in this work is intended to bypass constraint (1) without artificially altering the power distribution profile. The driving hypothesis is that for a long enough heating track, the thermal field will reach a steady state in the reference frame of the heat source. Once steady, larger time-steps are allowed. In the welding literature, multiple authors have observed that steady state is reached rather quickly (Mundra et al., 1996; Van Elsen et al., 2007; Powar et al., 2016). Computing the whole problem in the reference frame of the heat source is unwise since it introduces transitoriness in far away regions and at the boundaries of the build. Hence, a Chimera approach (Houzeaux et al., 2014) is applied: a patch mesh is attached to the laser and the rest of the domain is covered by a background mesh. The thermal field in the patch mesh is computed in the heat source reference frame while the background mesh does it in the laboratory reference

[3]Note that the potentially obtainable speedup factor for a given method is problem dependent.

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frame. The two meshes are coupled at their interface through Neumann-Dirichlet transmission conditions. This framework is similar to the one presented by Storti et al. (2022), the main difference being the coupling of subdomains.

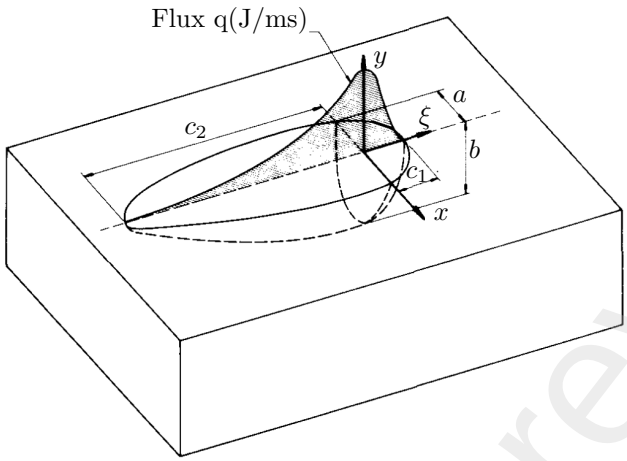


Figure 2: Double ellipsoidal profile (Goldak et al., 1984). ξ is the welding direction.

Furthermore, it will be observed that this approach offers accuracy gains in the HAZ. Goldak et al. (1984) noted that the temperature gradient obtained with an ellipsoidal heat source is inaccurate in the proximity of the meltpool. In its back, the gradients are too steep, while in its front, they are too flat. To remediate this issue, the double ellipsoidal heat source profile was proposed, which to this day is still widely used to conform to the final asymmetric shape of the meltpool: note on fig. 2 that a shorter ellipsoid will be placed in the front of the welding path and a longer one at the rear. Thus, the symmetrical Gaussian profile is artificially modified to account for the missing heat advection induced by the fast motion of the heat source. Moreover, the determination of the parameters characterizing the double ellipsoid is not straightforward (King et al., 2015). This work will show that using the heat source reference frame will naturally remediate the advection effect described by Goldak without requiring the use of an asymmetrical profile.

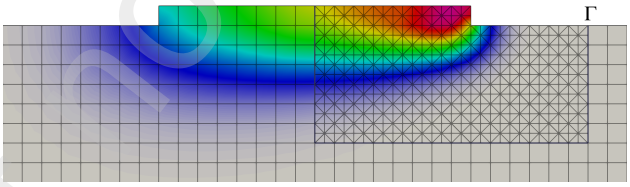


Figure 3: The subdomain of the HAZ can be meshed with a different element type, size or degree.

It is remarked that Chimera methods also allow for the use of a finer patch mesh compared to the background mesh (cf. fig. 3), avoiding the use of AMR in the HAZ and the associated time due to the recomputation of connectivities at every time-step (Storti et al., 2022; Carraturo et al., 2021).

2. Methodology

This section is structured as follows: first, the standard approach in the laboratory reference frame is presented. For the remainder of this work, it will be referred to as *reference model*. Then the Chimera method is described.

2.1. Reference model

The reference model in this work consists of the balance of energy equation for AM. For the sake of simplicity neither the thermal dependency of the material parameters nor the latent heat are accounted for. The problem statement is as follows: solve for the temperature field $T = f_{(x,t)} : \Omega(t) \mapsto \mathbb{R}$ such that

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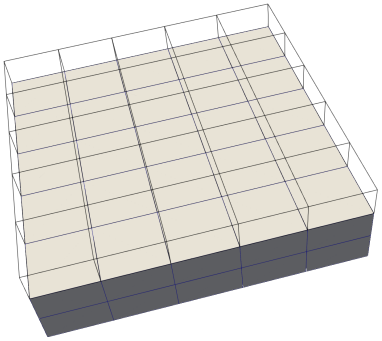
$$\begin{aligned} \rho c_p \partial_t T_{(x,t)} - \nabla \cdot (k \nabla T_{(x,t)}) &= r_{(x,t)} && \forall x \in \Omega(t) \\ q_{conv} &= h_{conv}(T_{(x,t)} - T_{\text{env}}) && \forall x \in \partial\Omega_{conv}(t) \\ T_{(x,0)} &= T_{0(x,0)} && \forall x \in \Omega(0) \end{aligned} \tag{2}$$

where ρ, c_p, k, r, q_{conv} and h_{conv} denote the density, the specific heat, the conductivity, the volumetric heat source, the convection heat flux and the convection coefficient, respectively. The heat source adopted in this work is the standard Gaussian distribution.

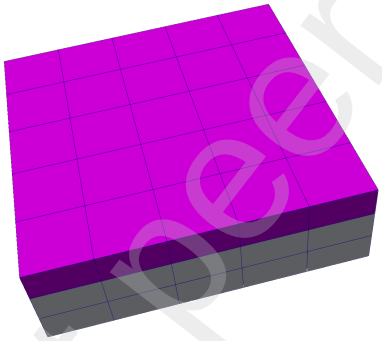
$$r_{2D(\mathbf{x},t)} = \frac{2P}{\pi R^2} \exp\left(\frac{-2||\mathbf{x} - \mathbf{x_{hs}}(t)||_2^2}{R^2}\right) \qquad r_{3D(\mathbf{x},t)} = \frac{6\sqrt{3}P}{\pi^{3/2}R^3} \exp\left(\frac{-3||\mathbf{x} - \mathbf{x_{hs}}(t)||_2^2}{R^2}\right) \tag{3}$$

where P, R and $\mathbf{x_{hs}}$ denote respectively the net power supply, the radius and the current position of the heat source. Note the time dependency of $\mathbf{x_{hs}}$ in the laboratory reference frame.

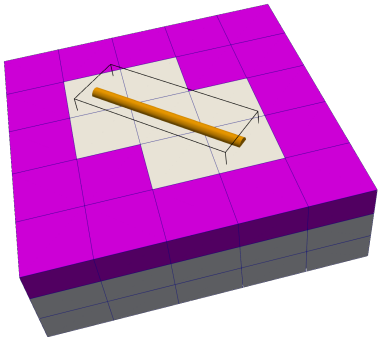
In this work, the powder bed is included in the simulation domain when considering LPBF applications. However, a surrogate model is used for the transformation of loose powder into bulk material. Elements affected by the HAZ are automatically set to bulk elements by increasing their conductivity by a factor of 100 as done in Hodge et al. (2014). Said elements are detected through a collision test as depicted in fig. 4.3.



(4.1) Bare substrate below an inactive layer.



(4.2) A powder layer is activated.



(4.3) The material of the elements which collide with the deposited hatch is set to bulk.

Figure 4: Illustration of simplified melting. ( ) Powder ( ) Bulk

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The time dependency of the computational domain $\Omega(t)$ in eq. (2) is due to the deposition process. Numerically, it is modelled using the born-dead element technique (Chiumenti et al., 2010). Elements of a layer that are yet to be deposited are *inactive* and are not included in the computational domain (cf. fig. 4.1). During the recoating step, a new layer of powder elements is activated as shown in fig. 4.2.

2.2. Chimera model

This section presents the Neumann-Dirichlet scheme employed in this work. In the context of this problem, it can be used for both overlapping and non-overlapping decomposition: the classic version is non-overlapping, and Houzeaux and Codina (2003) shows its validity in overlapping settings. For the sake of simplicity, the non-overlapping version is the one featured in the body of the work while the overlapping one is discussed shortly in the appendix.

In the vicinity of the HAZ, the problem is modelled in the reference frame of the moving heat source. To this end, a moving subdomain $\Omega_m(t) \subset \Omega$ is attached to the source, such that the latter appears fixed in space when observed from Ω_m. Let ξ denote the spatial coordinate in this frame and $\mathbf{v} = \frac{d\mathbf{x_{hs}}}{dt}$ the speed of the laser. The change of coordinate system writes

$$\xi = \mathbf{x} - \int_0^t \mathbf{v}_{(t)} dt$$

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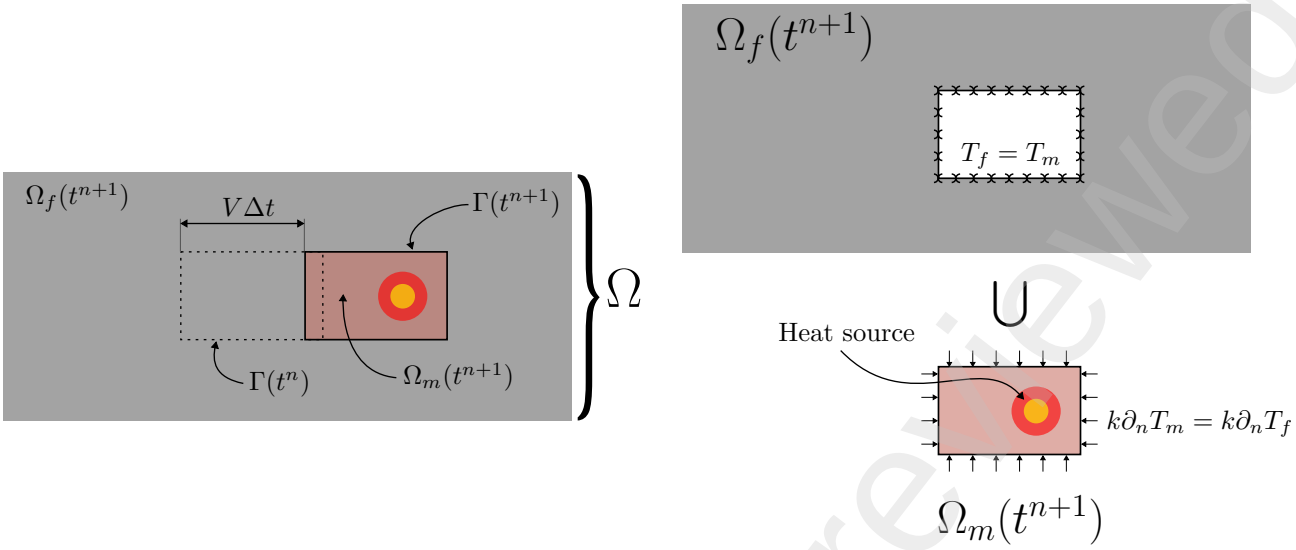


Figure 5: Schematic of Chimera model with non-overlapping decomposition.

In this framework, the balance of energy (2) can be rewritten for $T^m : \Omega_m \to \mathbb{R}$

$$\rho c_p \Big(\partial_t T^m{}_{(\xi,t)} - \mathbf{v}_{(t)} \cdot \nabla T^m{}_{(\xi,t)} \Big) - \nabla \cdot (k \nabla T^m{}_{(\xi,t)}) = r_{(\xi)} \qquad \xi \in \Omega_m(t) \tag{4}$$

Note the introduction of an advection term to account for the motion of the mesh. Moreover, the power source no longer depends on time.

The rest of the computational domain is *fixed* and denoted by $\Omega_f(t) = \mathrm{int}\big(\Omega \setminus \Omega_m(t)\big) \subset \Omega$. The fixed subproblem is posed in the usual laboratory reference frame. If Ω_m is big enough to contain the whole support of the source, the balance equation for Ω_f writes as follows for $T^f : \Omega_f \to \mathbb{R}$

$$\rho c_p \partial_t T^f{}_{(x,t)} - \nabla \cdot (k \nabla T^f{}_{(x,t)}) = 0 \qquad x \in \Omega_f(t) \tag{5}$$

Both subdomains form a decomposition of Ω at t^{n+1} and are coupled through Neumann-Dirichlet transmission conditions at their interface, $\Gamma(t) = \partial\Omega_m \cap \partial\Omega_f$, as shown on fig. 5.

$$k\partial_n T^m{}_{(\xi,t)} = k\partial_n T^f \qquad \xi \in \Gamma(t) \tag{6}$$

$$T^f{}_{(x,t)} = T^m \qquad x \in \Gamma(t) \tag{7}$$

The remaining heat transfer and initial conditions write

$$\begin{aligned}
q_{\mathrm{conv}} &= h_{\mathrm{conv}}(T^m{}_{(\xi,t)} - T_{\mathrm{env}}) && \xi \in \partial\Omega_{m,\mathrm{conv}}(t) \\
T_{(\xi,0)} &= T_{0(\xi,0)} && \xi \in \Omega_m(0) \\
q_{\mathrm{conv}} &= h_{\mathrm{conv}}(T^f{}_{(x,t)} - T_{\mathrm{env}}) && x \in \partial\Omega_{f,\mathrm{conv}}(t) \\
T_{(x,0)} &= T_{0(x,0)} && x \in \Omega_f(0)
\end{aligned}$$

At time t^{n+1}, Ω_m and Ω_f are discretized in space by the FE meshes $\mathcal{T}^m(t)$ and $\mathcal{T}^f(t)$. *Streamline Upwind Petrov-Galerkin* (SUPG) is chosen to stabilize the advection term in eq. (4). The resulting coupled integral equations in $T_h^m \in V_h^m$ and $T_h^f \in V_h^f$ write as follows

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$$\int_{\Omega_m} \phi_m \rho c_p \partial_t T_h^m - \int_{\Omega_m} \phi_m \rho c_p \mathbf{v} \cdot \nabla T_h^m \tag{8}$$
$$+ \int_{\Omega_m} k \nabla \phi_m \cdot \nabla T_h^m - \int_{\partial\Omega_{m,\text{conv}}} \phi_m h_{\text{conv}} T_h^m$$
$$- \int_\Gamma k \phi_m \nabla T_h^f \cdot \hat{n} + \int_{\Omega_m} \tau R_h\big|_{T_h^m} \rho \mathbf{v} \cdot \nabla \phi_m = \int_{\Omega_m} \phi_m r + \int_{\partial\Omega_{m,\text{conv}}} \phi_m h_{\text{conv}} T_{\text{env}} \qquad \forall \phi_m \in V_h^m$$

$$\int_{\Omega_f} \phi_f \rho c_p \partial_t T_h^f + \int_{\Omega_f} k \nabla \phi_f \cdot \nabla T_h^f - \int_{\partial\Omega_{f,\text{conv}}} \phi_f h_{\text{conv}} T_h^f = \int_{\Omega_f} \phi_f r + \int_{\partial\Omega_{f,\text{conv}}} \phi_f h_{\text{conv}} T_{\text{env}} \qquad \forall \phi_f \in V_h^f \tag{9}$$

$$V_h^m \subset H^1(\Omega_m) \qquad V_h^f \subset H^1(\Omega_f) \,:\, \forall \phi_f \in V_h^f,\ \phi_f\big|_\Gamma = 0$$

where $R_h\big|_{T_h^m}$ denotes the pointwise residual of T_h^m in equation eq. (4) and τ is an element-wise scalar stabilization coefficient chosen according to Codina (2000). Backward Euler is adopted for the time integration. Note that higher order time integrations such as Crank-Nicholson can introduce oscillations (see Puso and Hodge (2023)). After time discretization, the algebraic form of the problem writes

$$\begin{pmatrix} \frac{\tilde{M}_m}{\Delta t} + K_{m,m} & K_{m,m_\Gamma} & \cdot & \cdot \\ K_{m_\Gamma,m} & \frac{\tilde{M}_{m_\Gamma}}{\Delta t} + K_{m_\Gamma,m_\Gamma} & \cdot & N \\ \cdot & \cdot & \frac{\tilde{M}_f}{\Delta t} + K_{f,f} & K_{f,f_\Gamma} \\ \cdot & D & \cdot & -I \end{pmatrix} \begin{pmatrix} \mathrm{T}^{m,n+1} \\ \mathrm{T}^{m_\Gamma,n+1} \\ \mathrm{T}^{f,n+1} \\ \mathrm{T}^{f_\Gamma,n+1} \end{pmatrix} = \begin{pmatrix} F_m + \frac{\tilde{M}_m}{\Delta t}\mathrm{T}^{m,n} \\ F_{m_\Gamma} + \frac{\tilde{M}_{m_\Gamma}}{\Delta t}\mathrm{T}^{m_\Gamma,n} \\ F_f + \frac{\tilde{M}_f}{\Delta t}\mathrm{T}^{f,n} \\ \cdot \end{pmatrix} \tag{10}$$

On the left-hand side, the $\mathrm{T}^{\cdot,k}$-vectors contain the nodal values of the solutions at time t^k, the $\tilde{M}_\cdot$-matrices denote the modified mass matrices with an extra SUPG term associated to the time derivative, the $K_{\cdot,\cdot}$ matrices contain diffusion, advection and convection contributions from the nodes of the second subscript to the nodes of the first subscript, matrix N contains the Neumann contribution from the Γ-nodes of $\mathcal{T}^f$ to the Γ-nodes of $\mathcal{T}^m$, and the last row corresponds to the Dirichlet condition (7). D is such that, at time t^k,

$$\text{row}_i D \cdot \mathrm{T}^{m,k} = T_h^m\big(\xi_i, t^k\big) = T_h^m\big(\mathbf{x}_i - \int_0^{t^k} \mathbf{v} dt, t^k\big) \qquad \forall \mathbf{x}_i \mid \mathbf{x}_i \text{ is a } \Gamma\text{-node of } \mathcal{T}^f$$

and I is an identity matrix of the appropriate size. On the right-hand side, the F-vectors contain power density and convection contributions. Notice that, for a monolithic solve, the Dirichlet and Neumann conditions need to be assembled to the left-hand side of eq. (10) since they depend on the solution at t^{n+1}.

Matrix (10) is non-symmetric: $K_{m,m}, K_{m,m_\Gamma}$ and $K_{m_\Gamma,m}$ contain advection contributions and matrices N and D are unrelated.

In the numerical examples, the linear system (10) will be solved monolithically at each time-step. In contrast, the reference model only requires the solution of a Symmetric Positive Definite (SPD) matrix. The performance of the method could be enhanced by solving for Ω_f and Ω_m in a staggered fashion. Although the matrix of the moving subproblem would remain non-symmetric, the matrix of the fixed subproblem will be SPD.

As the moving subproblem grows steadier, larger Δt and Ω_m are chosen according to the following workflow

$$\texttt{steadiness_metric}(T_h^m) < \alpha \implies \text{Increase } \Delta t \implies \text{Resize } \Omega_m \tag{11}$$

where $\alpha > 0$ is an given threshold. This is checked at the end of each time-step, and Δt is generally increased by a constant value $0.5\,T_{hs} - 1T_{hs}$. The following ratio of L_2 norms is utilized:

$$\texttt{steadiness_metric}(f) := \frac{\left\| f\big|_{t^{n+1}} - f\big|_{t^n} \right\|_2}{\left\| f\big|_{t^{n+1}} \right\|_2} \tag{12}$$

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That is, the solution at t^n and t^{n+1} is used to determine whether Δt in $t^{n+2} = t^{n+1} + \Delta t$ can be increased. This metric does not take into account changes in the motion of the heat source, the material composition, nor the geometry. In this work, the former is taken care of by resetting Δt upon starting a new heating track, but the other factors mentioned are left unaddressed.

Regarding the shaping and sizing of Ω_m, the following considerations must be taken into account:

1. $\Omega_m \subset \Omega$. Since $\mathcal{T}^m$ is moving, it is necessary to check whether any elements went out of bounds. In particular, assuming a first order time integration, one needs to ensure that

$$\Omega_m(t^n) \subset \Omega(t^n) \tag{13}$$

$$\Omega_m(t^{n+1}) \subset \Omega(t^{n+1}) \tag{14}$$

   The first condition ensures that the previous solution is available, and the second condition is trivially necessary.
2. Ω_m should be big enough to contain the heat source so that it can be omitted in eq. (5). If a portion of its support is left out of Ω_m, the right-hand side of eq. (5) is non-zero and the spotty pattern of fig. 1.2 might reappear. For the 3D heat source of eq. (3), the minimum size of Ω_m should be about 1.5-2 radii, since, at that distance, the relative power density is negligible.
   Note that it is not always possible to guarantee this minimum size since condition (13) might interfere: consider the situation described in fig. 6 where the heat source at t^{n+1} is at a boundary of the domain and its speed is perpendicular to the normal i.e. $\mathcal{T}^m$ is coming onto the domain. Although subset B belongs to Ω, it can not be included in Ω_m since, at t^n, it was outside Ω and breaks (13). In this case, since B contains an important portion of the support of the heat source, the scheme is restricted to small time-steps, or it can revert to the reference model by engulfing all of Ω in Ω_f.
3. Ω_m should be long enough not to skip over a portion of Ω_f i.e. its back length must be bigger than $V\Delta t$. This is analogous to constraint eq. (1), with the advantage here that Ω_m can be sized at each time-step.
4. Ω_m should not include regions of the solution which are steadier in the laboratory reference frame e.g. boundaries that are not the deposition front.

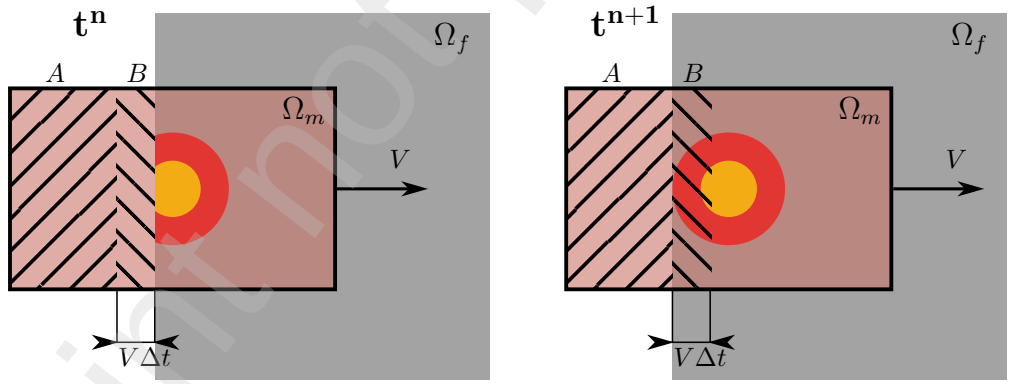


Figure 6: Schematic with Ω_m coming onto the domain. A and B violate conditions (14) and (13), respectively: at time t^n, B was out of bounds, hence the time derivative can not be computed on $B \subset \Omega_m$. Consequently, it has to be included in Ω_f.

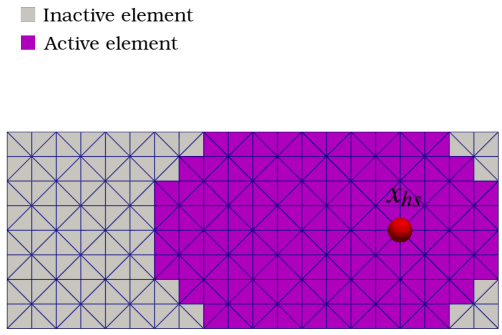
At the implementation level, Ω_m is sized according to the following parameters: front, back, side, top and bottom lengths with respect to the position of the heat source. The front, side, top and bottom lengths are usually set according to item 2. To set the back length, a function of Δt is provided to the software keeping in mind item 3. Item 1 is enforced at the level of the driver using boolean operations before and after the motion of Ω_m as will be explained shortly.

The necessary degree of flexibility in adapting Ω_m at each time-step is achieved by employing a mesh larger than necessary and deactivating elements until the desired shape and size are attained. This process is illustrated in fig. 7. Let $\mathcal{T}^m_{bg}$ denote this background mesh, such that $\mathcal{T}^m$ now refers to the active mesh of Ω_m. A time-step starts by resetting $\mathcal{T}^m$ and $\mathcal{T}^f$ to $\mathcal{T}^m_{bg}$ and a mesh of $\Omega(t^n)$, respectively. When a new

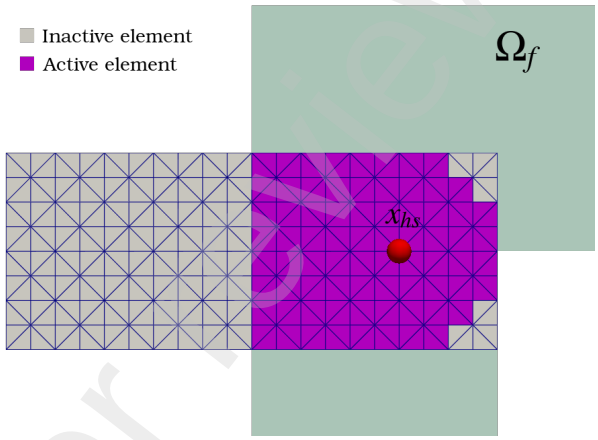
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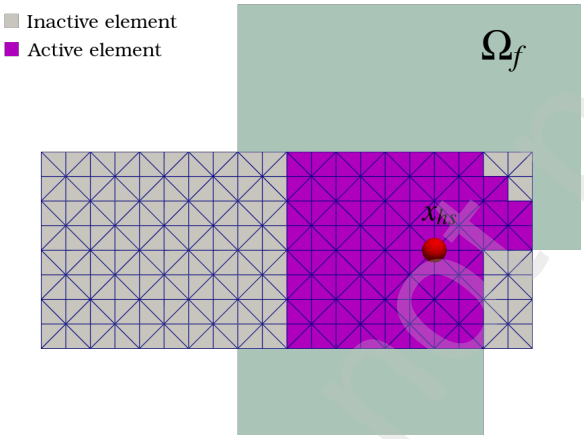
heating track starts, $\mathcal{T}_{bg}^m$ is aligned with the new heat source direction through a rotation. Then, $\mathcal{T}_{bg}^m$ is progressively reduced to $\mathcal{T}^m$ through multiple intersections. First, the routine `shapeSubdomain` sets the desired lengths by intersecting $\mathcal{T}_{bg}^m$ with oriented boxes or/and spheres of the corresponding sizes. Then, $\mathcal{T}^m \leftarrow \texttt{intersect}\left(\mathcal{T}^m, \mathcal{T}^f\right)$ is performed to enforce eq. (13). After the motion of $\mathcal{T}^m$ from t^n to t^{n+1}, $\mathcal{T}^m \leftarrow \texttt{intersect}\left(\mathcal{T}^m, \mathcal{T}^f\right)$ is carried out again so that eq. (14) is respected. Finally, the *hole cutting* operation is performed: the resulting mesh for $\mathcal{T}^m$ is subtracted from the mesh of $\mathcal{T}^f$, i.e. $\mathcal{T}^f \leftarrow \texttt{subtract}\left(\mathcal{T}^f, \mathcal{T}^m\right)$.



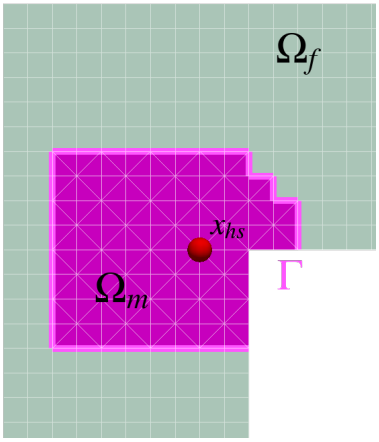
(7.1) $\mathcal{T}^m \leftarrow \texttt{shapeSubdomain}(\mathcal{T}_{bg}^m)$; the background mesh is intersected with boxes and spheres to impose the necessary lengths.



(7.2) $\mathcal{T}^m \leftarrow \texttt{intersect}\left(\mathcal{T}^m, \mathcal{T}^f \text{ at } t^n\right)$ in order to impose eq. (13).



(7.3) $\mathcal{T}^m \leftarrow \texttt{intersect}\left(\mathcal{T}^m, \mathcal{T}^f\right)$ at t^{n+1} after the subdomain motion in order to impose eq. (14).



(7.4) $\mathcal{T}^f \leftarrow \texttt{subtract}\left(\mathcal{T}^f, \mathcal{T}^m\right)$ and a non-overlapping decomposition is obtained.

Figure 7: Illustration of the meshing process for Ω_m and the boolean operations carried out during an iteration.

Activation is kept track of through the `activeElements` and `activeNodes` arrays. The aforementioned boolean operations are implemented by interpolating the `activeNodes` array as a finite element function from a mesh to another. At the receiving end of the projection, elements whose nodes are all 1-valued are marked as active in the other mesh. The values of the `activeElements` array are modified accordingly: for `intersect`, an element is kept active only if active in both meshes, and for `subtract`, an element is deactivated only if active in the other mesh. The latter can result in overlaps if the boundary of $\mathcal{T}_m$ lies in the middle of elements of $\mathcal{T}_f$. This case will be discussed in the appendix. In the numerical examples, only small overlaps will occur, since, after the hole cutting, no distance computations are performed to enlarge the overlap.

Algorithm 1 describes the driver routine of the Chimera method. Note that although the Chimera method adds a considerable amount of operations, most of these are projections and spatial searches whose cost is

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negligible compared to the assembly and the solution of the linear system[4]. As shown here, the driver routine would need to be modified for another choice of integration scheme to check, for instance, that the previous solution at t^{n-1} is available at nodes.

Algorithm 1 Driver routine for LPBF time-step of Chimera method
Additional operations with respect to reference method are highlighted.

$\mathcal{T}^m \longleftarrow \mathcal{T}^m_{bg}$ ▷ *Reset activation of* $\mathcal{T}^m$
$\mathcal{T}^f \longleftarrow$ Mesh of $\Omega(t^n)$ ▷ *Set* Ω_f *to physical domain*
if on new layer **then**
    Activate powder layer
if on new track **then**
    Set heat source power and speed ▷ *In* Ω_f
    Realign $\mathcal{T}^m$, set advection and domain speeds
$\mathcal{T}^m \leftarrow$ `shapeSubdomain`$(\mathcal{T}^m_{bg})$ ▷ *Shape* $\mathcal{T}^m$ *by intersecting its background mesh with boxes or/and spheres*
$\mathcal{T}^m \longleftarrow$ `intersect`$(\mathcal{T}^m, \mathcal{T}^f)$, ensuring $\Omega_m \subset \Omega(t^n)$
Advance time to t^{n+1} ▷ *Update heat source position, mesh motion, etc.*
$\mathcal{T}^m \longleftarrow$ `intersect`$(\mathcal{T}^m, \mathcal{T}^f)$, ensuring $\Omega_m \subset \Omega(t^{n+1})$
for all elements which collide with deposition **do**
    Set material to bulk ▷ *Simplified melting*
$\mathcal{T}^f \longleftarrow$ `subtract`$(\mathcal{T}^f, \mathcal{T}^m)$
Locate Γ_m and Γ_m ▷ *Locate the interface facets of both domains*
Set Γ_f to Dirichlet and Γ_m to Neumann boundary conditions
Set convection boundary conditions on external boundary ▷ $\partial\Omega \setminus \Gamma$, *necessary to update since the meshes are changing*
Assemble and solve the finite element problem
Communicate solution across meshes
Compute steadiness of moving problem and new Δt

3. Numerical examples

Both the reference and the Chimera methods were implemented in a `C++` library. All the code for both the algorithm and the cases run is made available online at the following link: `https://github.com/ordinary-slim/moving_heat_source`. The main routines are exposed to `Python` through `Pybind` (Jakob et al., 2017) in order to write the driver code in a high level language. Third party pre and post-processing libraries are invoked through their `Python` APIs (Application Programming Interface). Namely, the libraries `gmsh` and `meshio` are extensively used (Geuzaine and Remacle, 2009; Schlömer et al., 2018).

For the processing, both the `Eigen` library (Guennebaud et al., 2010) and `MATLAB` (Inc., 2023) are used for linear algebra, the later being invoked through its `C++` API. Geometrical operations (convex polyhedra collision, fast spatial search) are implemented using `CGAL` (Hert and Schirra, 2023; Alliez et al., 2023).

For each numerical example, both the reference and the proposed models described in section 2 are exposed. All the domains are meshed using first order quadrilateral and hexahedral elements.

The first example will showcase the accuracy of the proposed model in the vicinity of the heat source and illustrate a problem encountered by the Chimera model with large time-steps when Γ is ill-placed. The

[4]Section 3.3 contains a short discussion on the main costs of an iteration.

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second example features a slightly more complex heating track with the issues that come along it. Finally, the third example treats a 3D printing case, a thin wall printed using LPBF[5].

3.1. 3D track in infinite body

The performance of the Chimera model is studied in the context of a single track in an infinite medium. Analytical solutions of several 3D heat source profiles ranging from ellipsoidal to uniform distributions are available (Nguyen et al., 1999; Van Elsen et al., 2007). Throughout this work, Ti64 material is utilized alongside LPBF-like parameters. The data is reported in table 1.

Symbol	Parameter	Value	Unit
ρ	Density	4300	kg/m^3
c_p	Specific heat	700	J/kgK
k	Conductivity	10	W/mK
V	Speed V	1000	mm/s
R	Radius R	0.1	mm
P	Power	50	W
T_0/T^{dep}	Initial / Environment temperature	25	C°

Table 1: Material properties and process parameters for section 3.1.

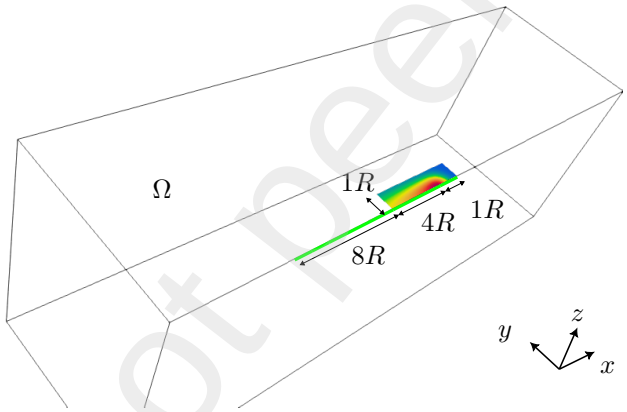


Figure 8: Computational domain Ω, subdomain of fig. 9 and line for profile of fig. 10.2.

The accuracy of the steady state solution in the proximity of the heat source as well as the effect of the interface Γ will be examined. The simulation domain Ω, although finite, is chosen big enough to neglect boundary effects in the subdomain of interest, cf. fig. 8. All the boundaries are insulated (null Neumann): the top and right surfaces are planes of odd symmetry of the solution and the remaining boundaries are assumed to be far enough so that the solution does not vary anymore.

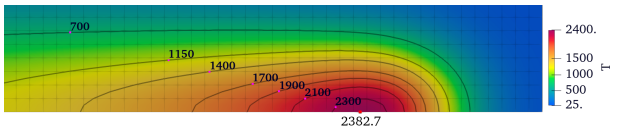
For the proposed model, a relatively large time-step of $\Delta t = 2\,T_{hs}$ is adopted in order to quickly reach steady state. In contrast, since the system is ever-transient in the reference model, it is necessary to use increasingly smaller time-steps in order to obtain convergence.

Figure 9 compares isothermal lines and peak temperatures of the analytical, the proposed and the reference models upon reaching quasi steady state. The reference model uses $\Delta t = 0.25\,T_{hs}$ while the proposed model employs the aforementioned $\Delta t = 2\,T_{hs}$. Both solutions are computed with a mesh density of 8 elements per radius. Note that the region shown is fully contained in Ω_m i.e. the temperature field of the

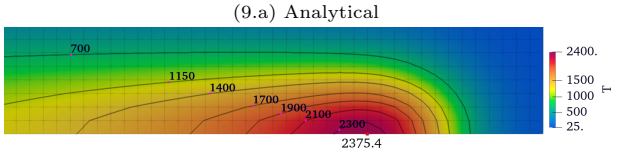
[5]It is worth mentioning that the authors tried applying the presented methodology to DED (Directed Energy Deposition) without success in reducing the total number of time-steps. The key difference with an LPBF simulation is that DED involves simultaneous heating and deposition at every printing step. Hence, at every time-step, the newly deposited nodes have a high time derivative of the solution and the moving subproblem is never steady. In discrete terms, upon activation of a node, T^n is set at the deposition temperature T^{dep}, resulting in $\Delta_h^t T = \frac{T^{n+1} - T^{dep}}{\Delta t} \gg 0$ in general.

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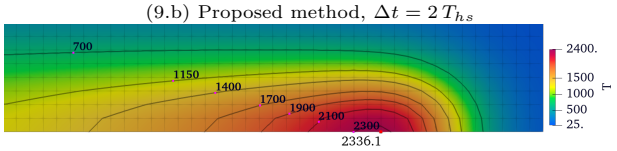
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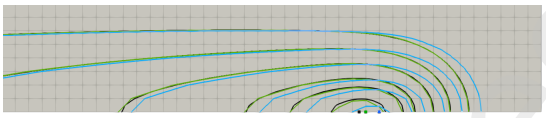
(9.a) Analytical



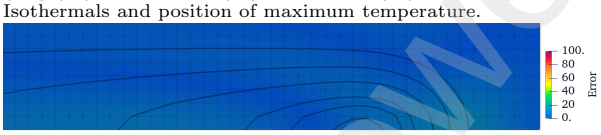
(9.b) Proposed method, $\Delta t = 2\,T_{hs}$



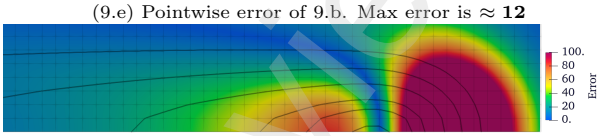
(9.c) Reference method, $\Delta t = 0.25\,T_{hs}$



(9.d) (■) Analytical (■) Proposed (■) Reference
Isothermals and position of maximum temperature.



(9.e) Pointwise error of 9.b. Max error is $\approx$ **12**



(9.f) Pointwise error of 9.c. Max error is $\approx$ **263**

Figure 9: Solution (left) and error (right) contour fields. Spatial resolution is 8 elements per radius.

proposed model is computed from the heat source reference frame. As expected, by virtue of being a better choice of reference frame, results are more accurate both in terms of peak temperature and isotherms. This is confirmed by table 2; at the finest spatial resolution, the solution in the heat source reference frame with $\Delta t = 2\,T_{hs}$ is still better than the solution in the laboratory reference frame with $\Delta t = \frac{1}{16}\,T_{hs}$. The latter is comparable to the 4 elements per radius solution in the heat source reference frame. Note from table 2b that, at a constant Δt, the error does not diminish with spatial refinement; indicating that it is solely affected by temporal refinement. Nonetheless, spatial refinement is still necessary in order to obtain convergence since the motion of the heat source must be resolved to avoid steps smaller than the size of an element: if the mesh has N elements per radius, $\Delta t \geq \frac{1}{N}\,T_{hs}$. Hence, convergence in the reference model needs both spatial and temporal refinement while it only involves spatial refinement in the proposed model due to it reaching steady state in Ω_m.

	$\Delta t = 2\,T_{hs}$
2 els. / radius	181.23%
4 els. / radius	44.73%
8 els. / radius	14.32%
16 els. / radius	5.46%

(a) **Proposed method**

	$\Delta t = 0.5\,T_{hs}$	$\Delta t = 0.25\mathcal{R}$	$\Delta t = 0.125\mathcal{R}$	$\Delta t = 0.0625\mathcal{R}$
2 els. / radius	403.41%	-	-	-
4 els. / radius	414.16%	208.76%	-	-
8 els. / radius	427.43%	216.71%	108.57%	-
16 els. / radius	431.37%	219.83%	110.99%	55.44%

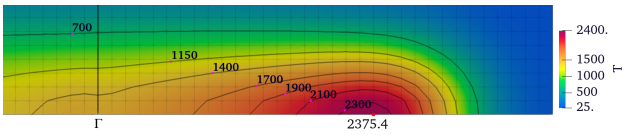
(b) **Reference method**

Table 2: Relative L2 error $\frac{\left\| T_h - T_{\text{anal}} \right\|_2}{\left\| T_{\text{anal}} \right\|_2}$ for 3D welding case.

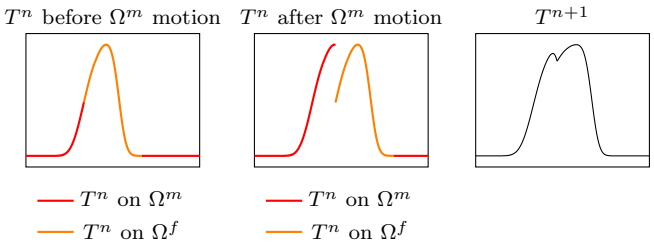
The error is concentrated at the thermal front since the steepest gradients are located there. The pointwise error there is particularly high in fig. 9.f because the predicted isotherms are too spread as illustrated on fig. 9.d, resulting in an underprediction of the gradients along the heating track. Conversely, in the thermal tail, the isotherms are too close, and the gradients are overpredicted. This issue is well known in the literature; Goldak et al. (1984) proposed an asymmetrical double ellipsoidal heat source with a smaller ellipsoid at the front and a longer ellipsoid at the back in order to compensate for these over- and underpredictions. Goldak suggests that the dimensions of these ellipsoids should correspond to the melt pool. Van Elsen et al. (2007) argues that such a heat source is artificial since physical heat sources are symmetrical. Note that this issue is absent in the Chimera method due to the advection, which, in the first place, was the one introducing the asymmetry: it brings closer the isotherms at the front and further apart the isotherms at the back by adding a thermal flux in the direction opposite to the laser speed. Concerning the peak temperature, the reference model predicts its position too far to the front. The Chimera model is again more accurate thanks to the presence of advection which pushes it further back.

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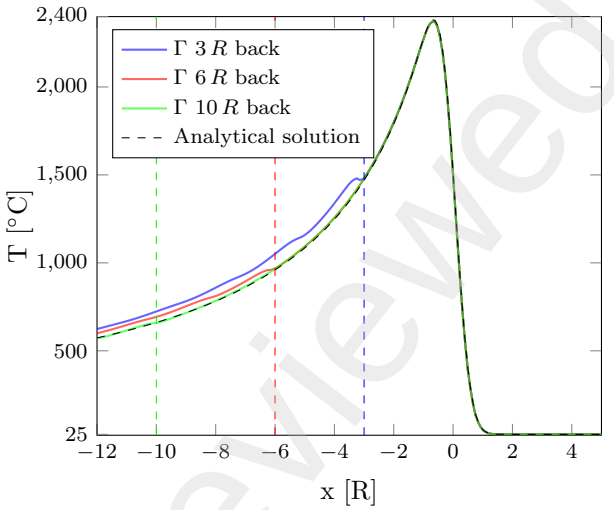
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(10.1) Proposed model, Γ 3R behind the heat source.



(10.3) Illustration of T^n discontinuity at $t^n n + 1$ resulting in oscillation.



(10.2) Temperature profiles along line shown on fig. 8 for Chimera model and interface positions progressively closer to the heat source, which is is located at 0. Distance is measured in radii.

In order to study the effect of the interface Γ, its position is brought much closer to the heat source as shown on fig. 10.1. Here, Γ is situated $3\,R$ behind the heat source, while keeping the same time-step $\Delta t = 2\,T_{hs}$. This change introduces oscillations as can be observed on the figure when the isotherm crosses Γ. This is a discretization effect due to the interface being located in a region with high gradients and the time-step being relatively too large[6]. The issue is expanded upon by fig. 10.2 which shows profile plots along the heating track over the line shown on fig. 8 for interface positions of $3\,R$, $6\,R$ and $10\,R$ behind the heat source. The $3\,R$ profile shows the highest oscillation, and, the further back the interface is positioned, the smaller the oscillation. The cause is depicted on fig. 10.3: T_h^n exhibits a spatial discontinuity across Γ after the motion of Ω_m. The moving subproblem carries over T_h^n of its previous position, discarding T_h^n from Ω_f at its new position which will not be used in the solution. The discontinuity in T_h^n causes a discontinuity in the discrete time derivative $\Delta_t T_h = \frac{T_h^{n+1} - T_h^n}{\Delta t}$ across Γ, and an oscillation of T_h^{n+1}. In 1D, the discontinuity writes

$$[\![T_h^n]\!]_{\Gamma^{n+1}} = \int_{x|_{\Gamma^n}}^{x|_{\Gamma^{n+1}}} \partial_x T_h^{f,n} dx = \int_{x|_{\Gamma^n}}^{x|_{\Gamma^n} + V\Delta t} \partial_x T_h^{f,n} dx \tag{15}$$

It is proportional to Δt and to the norm of the gradients in the region. If small enough, the discontinuity will be absorbed by diffusion as shown on fig. 10.2. The oscillation is also proportional to the speed of the heat source seeing as, for higher speeds, lower time-steps are required to advance the same distance, limiting the time over which the discontinuity can dissipate. Hence, the oscillation is proportional to the Péclet number. In the context of our problem, once the thermal tail is formed, it is advantageous to place the interface far behind as shown in fig. 10.2 so as to position Γ where gradients are low.

This oscillation could also be interpreted as a violation of the flux continuity condition in eq. (6), which is only weakly enforced. The issue could potentially be addressed by using a Lagrange multiplier approach which would allow giving bigger weight to eq. (6).

3.2. 2D square track

A 2D square welding track is featured here to show an example with sharp direction changes. The domain and the heating track are depicted on fig. 11. The material parameters are the same as in table 1. The speed is set to 100 *mm/s*, the radius to 0.2 *mm* and the power to 100 *W*. A convection boundary condition is chosen with a coefficient of 20 W/m and $T_{env} = 25$.

[6]To the knowledge of the authors, this effect has not been studied previously.

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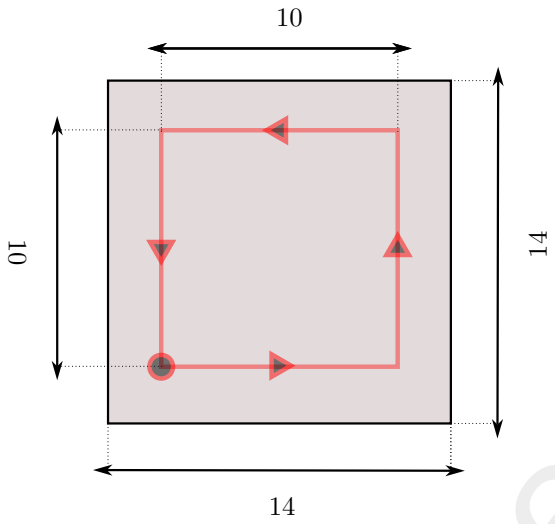


Figure 11: Domain and heating track for section 3.2.

The spatial resolution of both models is 4 elements per radius. The reference model uses $\Delta t = 0.5\,T_{hs}$ while the Chimera model progressively increases from $0.5\,T_{hs}$ to $5.5\,T_{hs}$.

Snapshots of the thermal fields towards the end of the simulation are shown on fig. 12. The reference method needs 400 time-steps, while the Chimera method achieves qualitatively similar results in 129 time-steps. The shape of Ω_m in fig. 12 is obtained by intersecting the rectangular background mesh with 2 circles and a rectangle following the procedure described in section 2. Note that the elongated rectangle has to be realigned through a rotation at the beginning of each track.

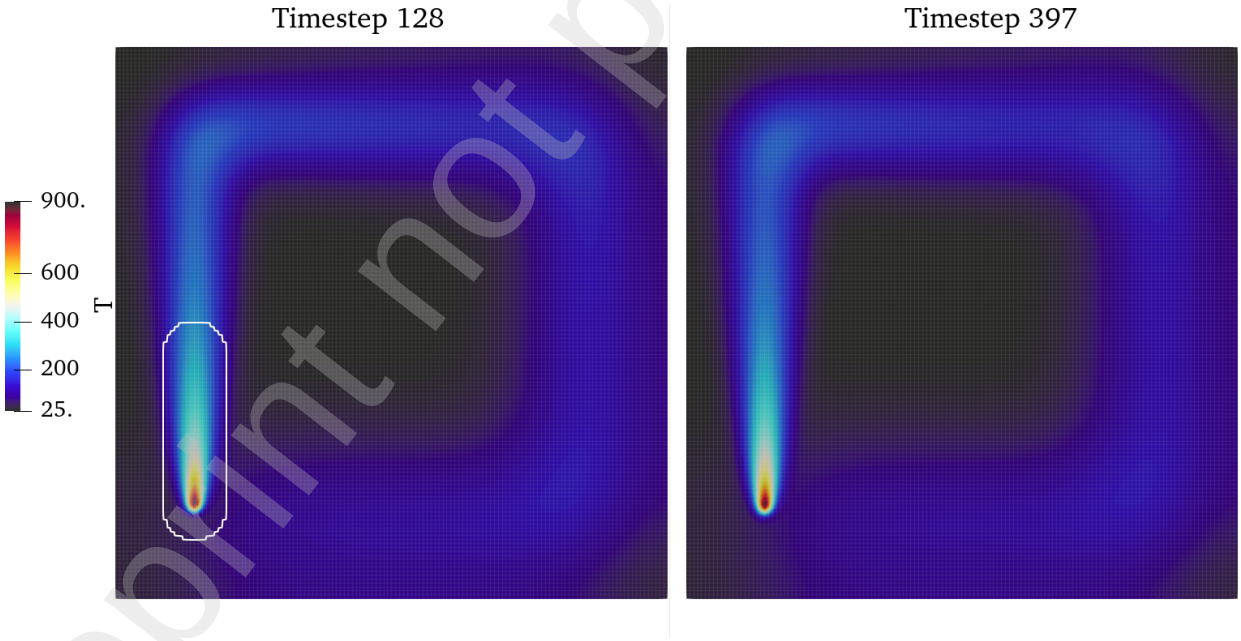


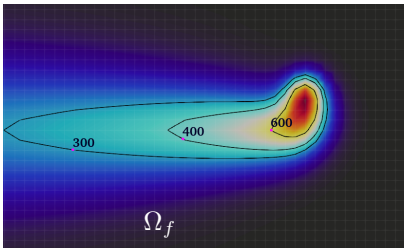
Figure 12: Proposed and reference method towards the end of the square track. The outline of Ω_m is shown in white after a time-step of $\Delta t = 5\,T_{hs}$.

Figure 13 zooms into the bottom right turn. The issue with sharp direction changes is depicted in fig. 13.2: the advection of the current heating track shears the previous thermal tail. Note that as the time-step is decreased, so does the shearing, as illustrated by fig. 13.3. The approach adopted to overcome this issue in the rest of this work is to turn off the scheme at the beginning of a heating track i.e. revert to the reference

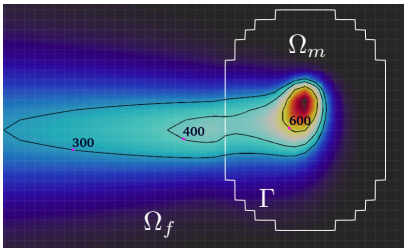
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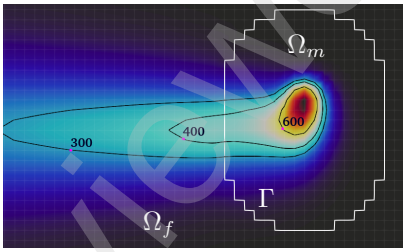
model. In fig. 13.1, right after a turn, Ω_f covers the whole domain. In the next time-steps, the solution will be projected on Ω_m, and whenever eq. (11) is satisfied for the first time along a track, the coupling will be turned back on. It is worth mentioning that upon turning the coupling back on, the temperature peak overshoots by around $5-10\%$ and goes back down after a couple of time-steps.



(13.1) Coupling turned off, $\Delta t = 0.5\,T_{hs}$.



(13.2) Coupling turned on, $\Delta t = 0.5\,T_{hs}$.



(13.3) Coupling turned on, $\Delta t = 0.25\,T_{hs}$.

Figure 13: Bottom right 90° turn. Γ is outlined in white. If coupling is on during sharp turns, shearing of the previous thermal tail is observed on the right.

The homepage of the code repository showcases a video of this example.

3.3. LPBF thin wall

The build of a LPBF thin wall is modelled with both the reference and the Chimera methods. The material and process parameters of table 1 are reused in this section. The wall is $10\,mm$ in length, $0.1\,mm$ in width and $0.125\,mm$ in height and the layer thickness is $25\,\mu m$ i.e. the simulation spans 10 layers. Each layer consists of a single track, whose direction alternates from one layer to the next. It is built on top of a $12\,mm \times 2\,mm \times 1\,mm$ substrate. Heat is convected across the boundary of the domain with $h_{conv} = 20\ W/m^2$ and $T_{env} = 25$. Interlayer cooling time is set to two seconds, after which the build is recoated by activating a new powder layer, as depicted on fig. 4.2.

Three solutions are compared: a solution with the Chimera model with increasing time-steps, a reference model solution with a standard $\Delta t = 0.5\,T_{hs}$ and a fine reference model solution with $\Delta t = 0.125\,T_{hs}$. The first two have a spatial resolution of 4 elements per radius in-plane and 1 element through the layer thickness. The fine solution has double the aforementioned resolution. The total number of DOFs is approximately the same for the first two solutions: it is slightly larger for the Chimera method since the interface nodes are duplicated. Figure 14.1 shows their mesh and bulk-powder distribution at the final time-step. The mesh shown has 293 K elements while the mesh of the fine solution has 1.891 M elements.

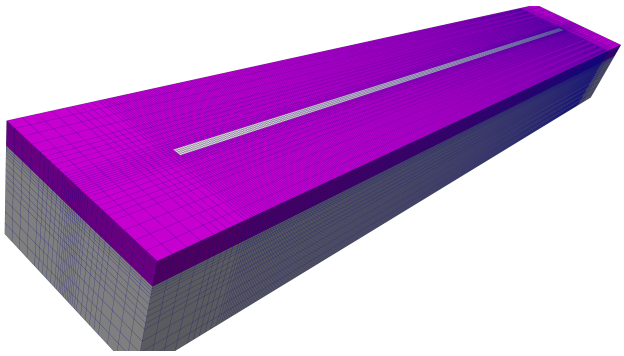
The time-step of the Chimera starts off at $0.5\,T_{hs}$ and increases in increments of $0.5\,T_{hs}$ until reaching $8\,T_{hs}$ towards the end of the track. As in the previous section, the coupling is turned off as appropriate: cooling time-steps and starts of tracks are modelled entirely from the laboratory reference frame i.e. the first heating time-steps of a track are solved in Ω_f and the solution is projected at each time-step to Ω_m. Similarly to section 3.2, the coupling is turned back on the first time that the solution in Ω_m satisfies eq. (12), meaning here the first time-step with $\Delta t = 1\,T_{hs}$.

The back length of Ω_m is chosen rather large so that the subdomain extends well into the thermal tail. This is done in order to mitigate oscillations as explained in section 3.1. For $\Delta t = 3\,T_{hs}$ and $8\,T_{hs}$, the back length is set to $10.25R$ and $50R$, respectively. Figure 14.2 shows the outline of Ω_m after a time-step with $\Delta t = 8\,T_{hs}$.

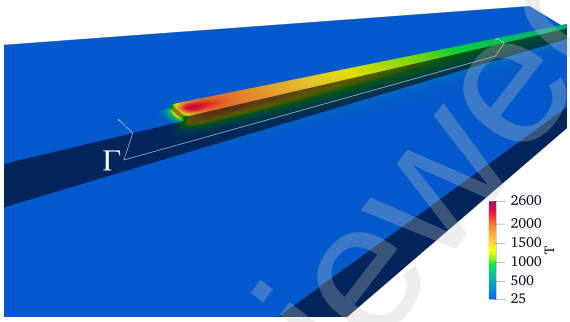
Figure 15 compares contours and profiles of the solutions. Figures 15.1, 15.4 and 15.5 display snapshots towards the end of the fifth layer while figs. 15.2, 15.3 and 15.6 show the end of the last track. For the proposed model, the size of the last time-step taken was $8\,T_{hs}$ and $1\,T_{hs}$ in fig. 15.5 and fig. 15.6, respectively. Note that the back of Ω_m is out of frame in the former while it is in frame in the latter. Oscillations are absent from fig. 15.5 and fig. 15.6 due to the appropiate choice of time-step & back length combination. The trend observed in the section 3.1 repeats here: isotherms obtained with the Chimera model are denser at the thermal front and looser at the tail, generally closer to the fine solution than the $0.5\,T_{hs}$-solution. In terms of peak temperature, both the Chimera and $0.5\,T_{hs}$ solutions are off by 30-80 C°.

Figures 15.7 and 15.8 show temperature profiles along the centerline at the top of the build. Similarly to the previous figures, they are timed towards the end of the fifth and tenth layers, respectively. Note that

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(14.1) Mesh and bulk-powder distribution at final time-step. ( ) Powder ( ) Bulk



(14.2) Temperature contours of the Chimera method on the metal component at an intermediate time-step. Γ is set 50 radii behind the heat source.

at $x = -5$ and $x = 5$, the material transitions from bulk to powder at the beginning/end of each hatch. In comparison to the $0.5\,T_{hs}$-solution, the Chimera one is slightly closer to the fine solution in the proximity of the HAZ but it is off by $30\,C^\circ$ in the back of the thermal tail. Slight oscillations are observable at the previous positions of Γ, especially on fig. 15.7 due to the issue explained in section 3.1. Note that only the fine solution is able to resolve the temperature spike at the bulk-powder interface.

Table 3 reports simulation costs in terms of time-steps and elapsed time for the Chimera and the $0.5\,T_{hs}$ solutions. Both methods are using `MATLAB`'s linear system solver. The Chimera method takes around 6 times less time steps and about 3.3 times less execution time. Note that the average Chimera iteration is around 2 times more expensive than the average iteration from the reference model despite the similar number of DOFs. This stems from the SPD-ness of the matrix of the reference model, which is a big advantage compared to the asymmetric monolithic matrix in (10).

The main costs of an iteration were studied on a similar case with 200 K DOFs. For a Chimera iteration, the solution of the system takes around 70% of an iteration, the assembly 19% and the writing of the results 7%. These costs were respectively $47\%, 41\%$ and 6% for a time-step in the reference method. The additional operations described in algorithm 1 take little time in comparison.

	Reference method, $\Delta t = 0.5\,T_{hs}$	Proposed method, adaptive Δt
Total printing time-steps	2000	331
Total duration printing time-steps [s]	22733	6832
Total decomposition time-steps	-	251
Average number of DOFs	258122	264168
Average duration printing time-step [s]	11.37	20.64
Average duration decomposition time-step [s]	-	23.32

Table 3: Cost of simulation in terms of number of time-steps and elapsed time for reference and Chimera methods. The solution of matrix equations is performed by `MATLAB`.

4. Conclusions

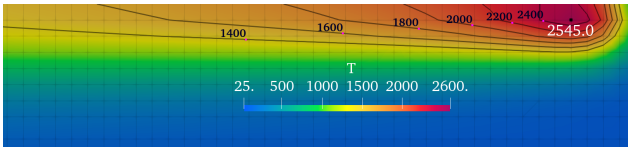
This works presents a novel FEM Chimera approach for the thermal modelling of moving heat sources. The problem in the HAZ is posed in the reference frame of the heat source while the rest of the domain is solved in the laboratory reference frame.

The main pros of the approach are an accuracy improvement in the HAZ and an increase in steadiness which can be exploited to take bigger time-steps since the formulation is not constrained by eq. (1). Both are due to the better choice of reference frame. In particular, the advection in the moving frame recovers a shorter front and a longer tail without requiring the use of an asymmetrical profile (Goldak et al., 1984).

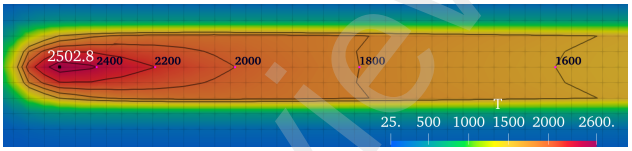
Its main cons are a higher algebraic complexity compared to the standard approach and the potential presence of oscillations in the solution depending on the choice of parameters and time-step. Concerning the former, advection loses the SPD-ness of the original problem and leads to a higher computational cost as

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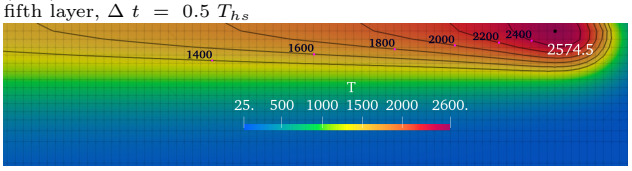
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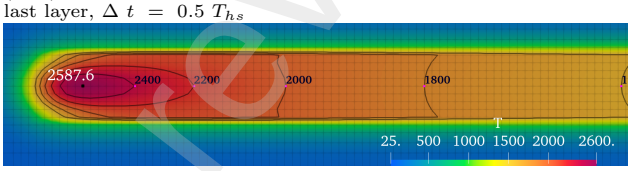
(15.1) Side view of reference model solution towards the end of the fifth layer, $\Delta\ t\ =\ 0.5\ T_{hs}$



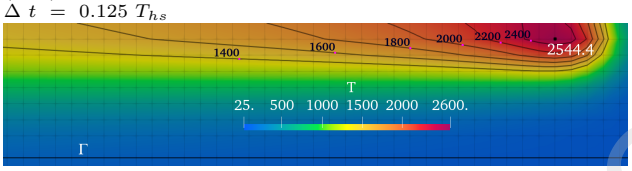
(15.2) Bird view of reference model solution towards the end of the last layer, $\Delta\ t\ =\ 0.5\ T_{hs}$



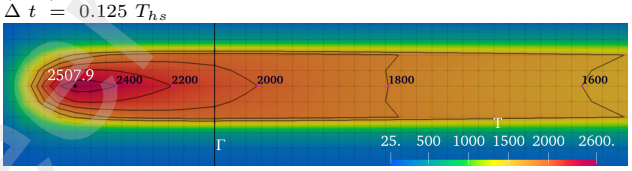
(15.3) Side view of fine solution towards the end of the fifth layer, $\Delta\ t\ =\ 0.125\ T_{hs}$



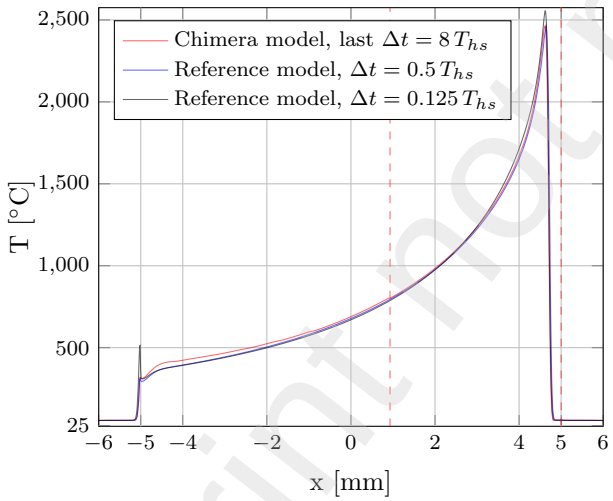
(15.4) Bird view of fine solution towards the end of the last layer, $\Delta\ t\ =\ 0.125\ T_{hs}$



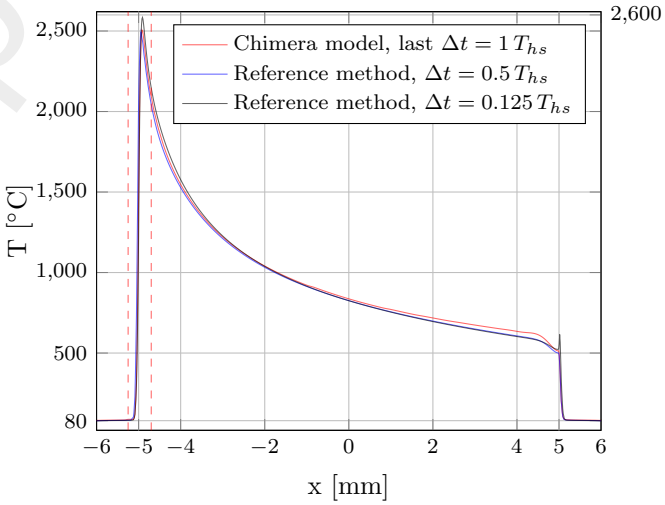
(15.5) Side view of Chimera model solution towards the end of the fifth layer, last $\Delta\ t\ =\ 8\ T_{hs}$



(15.6) Bird view of Chimera model solution towards the end of the last layer, last $\Delta\ t\ =\ 1\ T_{hs}$



(15.7) Plot along top centerline of build towards end of fifth layer.



(15.8) Plot along top centerline of build towards end of tenth layer.

Figure 15: Contours, peak temperatures and profiles along printing direction for intermediate (left, side view) and final times (right, bird view).

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described in section 3.3. Regarding the latter, formulations with moving subdomains present a discontinuity in the solution at the previous time-step. If this discontinuity is big enough, it can result in an oscillation as described in section 3.1. In the context of the problem at hand, this discontinuity can be minimized by optimizing the position of Γ at the thermal tail. A possibility is a spatial search in order to minimize a multidimensional version of eq. (15).

The authors believe that this approach can be combined with a substepping algorithm (Puso and Hodge, 2023; Viguerie et al., 2022) in order to further reduce simulation cost while retaining the aforementioned pros. Future works could also explore other choices of domain decomposition frameworks (Storti et al., 2022; Carraturo et al., 2021).

5. Acknowledgments

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Appendix A. Overlapping decomposition

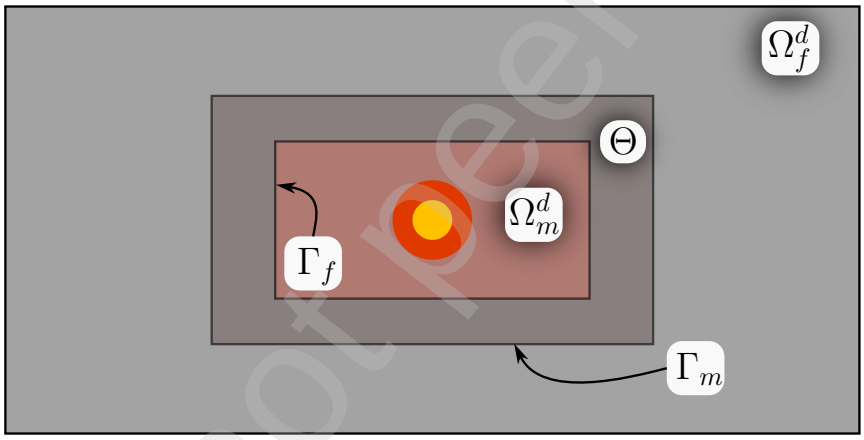


Figure A.16: Overlapping decomposition of Ω

Section 2 describes a non-overlapping Neumann-Dirichlet domain decomposition scheme. For the problem at hand, the same scheme is also valid in an overlapping setting (Houzeaux and Codina, 2003; Houzeaux, 2002). Let Ω_m^d, Ω_f^d and Θ denote the disjoint parts of the moving and fixed domains and the overlap, respectively. The decomposition writes

$$\begin{aligned} \Omega &= \mathrm{int}\left(\overline{\Omega_m^d \cup \Omega_f^d \cup \Theta}\right) \\ \Omega_m &= \Omega_m^d \cup \Theta \\ \Omega_f &= \Omega_f^d \cup \Theta \end{aligned}$$

with interfaces Γ_m and Γ_f as depicted on fig. A.16. For this decomposition, the system of differential equations becomes

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$$
\begin{aligned}
\rho c_p\Big(\partial_t T^m{}_{(\xi,t)} - \mathbf{v}_{(t)} \cdot \nabla T^m{}_{(\xi,t)}\Big) - \nabla \cdot (k \nabla T^m{}_{(\xi,t)}) &= r_{(\xi)} && \xi \in \Omega_m \qquad \text{(A.1)}\\
\rho c_p \partial_t T^f{}_{(x,t)} - \nabla \cdot (k \nabla T^f{}_{(x,t)}) &= 0 && x \in \Omega_f \\
k \partial_n T^m{}_{(\xi,t)} &= k \partial_n T^f && \xi \in \Gamma_m \\
T^f{}_{(x,t)} &= T^m && x \in \Gamma_f \\
q_{\text{conv}} &= h_{\text{conv}}(T^m{}_{(\xi,t)} - T_{\text{env}}) && \xi \in \partial\Omega_{m,\text{conv}} \\
T_{(\xi,0)} &= T_{0(\xi,0)} && \xi \in \Omega_m\big|_{t=0} \\
q_{\text{conv}} &= h_{\text{conv}}(T^f{}_{(x,t)} - T_{\text{env}}) && x \in \partial\Omega_{f,\text{conv}} \\
T_{(x,0)} &= T_{0(x,0)} && x \in \Omega_f\big|_{t=0}
\end{aligned}
$$

where the only difference lies in the transmission conditions being imposed at Γ_m and Γ_f instead of Γ. In its weak form, this domain decomposition scheme is the same as the one in proposition 4.1 of Houzeaux (2002). Although at a first glance they might appear different since the aforementioned is homogeneous, the heterogeneity is taken care of by expressing both PDEs in the same reference frame.

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